Subject: Date:

given data

	fit	unfit	Risky
fit	6	2	0
unfit	1	2	1
Risky	0	2	2

total = 6+2+0+1+2+1+0+2+2
= 16

~~for fit class~~

TP = 6

FP = 1+0 = 1

FN = 2+0 = 2

TN = ~~tot~~ − TP − FP − FN
= 16 − 6 − 1 − 2
= 7

fit − Calculation

Precision $= \frac{TP}{TP+FP} = \frac{6}{6+1} = 6/7 = 0.857$

Recall $= \frac{TP}{TP+FN} = \frac{6}{6+2} = \frac{6}{8} = 0.75$

specificity $= \frac{TN}{TN+FP} = \frac{7}{7+1} = 7/8 = 0.875$

f_score $= \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 * 0.857 * 0.75}{0.857 + 0.75} = 0.799$

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For unit class

TP = 2

FP = 2+2 = 4

FN = 1+1 = 2

TN = tot − TP − FP − FN

= 16 − 2 − 4 − 2

= 8

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+4} = 0.333$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+2} = 0.5$$

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{8}{8+4} = 0.667$$

$$\text{F-score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 * (0.333 * 0.5)}{0.333 + 0.5} = 0.399$$

Subject:

Subject:

Date:

For risk Class:

$TP = 2$

$FP = 0+1 = 1$

$FN = 2$

$TN = tot - TP - FP - FN$

$= 16 - 2 - 1 - 2$

$= 11$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+1} = 2/3 = 0.667$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+2} = 2/4 = 0.5$$

$$\text{specificity} = \frac{TN}{TN+FP} = \frac{11}{11+1} = 11/12 = 0.9167$$

$$F_1\text{-score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2 * (0.667 * 0.5)}{0.667 + 0.5}$$

$$= 0.57$$

$$\text{Accu} = \frac{\Sigma TP}{tot} = \frac{6+2+2}{16}$$

$$= 10/16 = 0.625 = 62.5\%$$